

Social Virtual Reality Is My Therapist: Overcoming Social Anxiety Disorder Through Using Social Virtual Reality

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ABSTRACT

Virtual Reality (VR) has long been an important research focus in HCI for treating different phobias. More recently, we have seen a more noticeable upgrade in VR, especially social VR. This research explores how socially anxious people use social VR to interact and practice social skills. The study analyzed 351 Reddit posts and comments and found that the immersive and embodied features of social VR, such as physicality, sense of presence, nonverbal communication and shared activities, can help individuals with a social anxiety disorder (SAD) overcome their fear of social interaction. Additionally, the study investigated if social skills gained in social VR can transfer to the offline world. This work highlights the potential of social VR in treating SAD and can inform the design of future social VR platforms to better support individuals with SAD.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in collaborative and social computing.**

KEYWORDS

Social VR, Social Anxiety, Social Phobia, Fear of Social interaction

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1 INTRODUCTION

Social interaction is essential for a broad range of life activities, including academic attainment, career development, emotional connections, and marriage. Unfortunately, SAD is a common concern among the young population, which can cause disruptions in social relations and have detrimental effects on an individual's overall well-being [18, 35]. Clinical research has consistently demonstrated that exposure therapy is the most effective treatment for SAD [11, 27]. This therapy can be conducted offline or in VR by exposing individuals to feared environments or situations. However, with VR exposure therapy (VRET), unlike offline exposure

therapy, the therapist has more control over the situation. Despite this advantage, the implementation of various virtual situations is costly and time-consuming [8, 10]. Individuals with SAD often use communication technologies like social networks and online gaming to make up for the lack of in-person social interaction in the offline world [5, 13, 17, 23, 23, 34]. However, previous research has shown that excessive use of traditional online communication platforms, such as Facebook, can negatively impact physical and psychological well-being and exacerbate social anxiety due to the lack of face-to-face communication [6, 13, 17, 28].

This paper aims to investigate the potential benefits of social VR for individuals with SAD in terms of social interaction. Social VR refers to a 3D virtual ecosystem where multiple users can interact and socialize through head-mounted displays [20, 22]. In this virtual environment, users can interact with each other and engage in a wide range of activities through embodied avatars, creating a highly realistic experience similar to face-to-face interaction. Popular social VR platforms include VRchat¹, Rec Room², Bigscreen³, AltspaceVR⁴, Meta Horizon⁵, and so forth. They tend to afford diverse activities and social atmospheres. For example, Rec Room focuses on VR gaming; VR Chat supports a wide range of creative activities and avatar customization; AltspaceVR is well known for its diverse event and professional development, and High Fidelity VR highlights large-scale public events and performance [22]. The potential use of this emerging virtual social space for treating social anxiety is acknowledged [4, 19, 21, 30], but there is limited understanding of how do socially anxious individuals may use social VR to support their lack of social interaction. To address this gap, the study employs a qualitative approach, analyzing 351 rich Reddit posts and comments about social VR users' experiences of using social VR to have social interaction in their life and practice social skills. The study aims to answer three research questions: (1) how do socially anxious individuals use social VR for social communication? (2) What strategies in social VR users use to mitigate social anxiety? (3) Would virtual socializing practiced through social VR will affect users' offline-world communication?

2 SOCIAL ANXIETY DISORDER, AND SOCIAL VIRTUAL REALITY

SAD is a prevalent mental illness characterized by fear of social situations due to potentially negative evaluations and scrutiny from others [2, 14]. Individuals with SAD often experience concerns such

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¹<https://hello.vrchat.com>

²<https://recroom.com>

³<https://www.bigscreenvr.com>

⁴<https://altvr.com>

⁵<https://www.meta.com/horizon-worlds/>

as shaking, blushing and appearing anxious during social communication [32]. Social relationships are crucial, and disruptions in these relationships can affect various aspects of life [18, 35]. A study found that SAD is a concern among young people, with 1 in 3 (36%) meeting the threshold criteria for the disorder [12].

In the treatment of SAD, exposure therapy is a widely used approach that aims to reduce fear reactions by exposing individuals to specific situations that cause anxiety [11, 27]. Despite its effectiveness, exposure therapy has some challenges, such as recreating the interpersonal situation that evokes the appropriate level of fear, difficulty imagining the feared situation, and lack of control over the situation [29]. Additionally, people with SAD have been reported to have barriers to treatment, such as uncertainty about where to go for treatment, concerns about what others might think or say, and financial constraints [25].

A significant body of research has investigated the impact of computer-mediated communication (CMC) and online gaming on SAD [5, 13, 17, 23, 34]. Due to the absence of physical and auditory cues, individuals with SAD often perceive the internet as a safe environment [23, 26]. On the one hand, CMC can provide a sense of anonymity through usernames in social networks, or avatars in massively multiplayer online role-playing games (MMORPGs) [16, 36] which help them conceal their real appearance and make a safe space for them to form social networks and receive support [3, 24, 34]. However, on the other hand, excessive use of internet-based communication platforms can exacerbate symptoms of SAD due to a lack of face-to-face interaction [6], such as problematic use of Facebook [17] or Instagram [13]. Although CMC may reduce social anxiety for some individuals, excessive use of that exacerbates symptoms of SAD.

In recent years, VRET has been found to be effective in addressing the challenges of traditional exposure therapy for SAD. It allows for the simulation of real-world situations relevant to the treatment context in VR, which provides a safe and controlled environment for practicing social skills under the guidance of a therapist [8, 9, 15]. While VRET can serve as an alternative to traditional exposure therapy, challenges such as the need to implement a variety of virtual situations that enable verbal interactions [8], the extent of efficacy, duration of effectiveness, and impact on therapy discontinuation [10] still remain.

Prior research has revealed that social VR can help individuals with SAD with their social needs [4, 19, 21, 30]. However, those studies have not focused on social anxiety, and less is known about how socially anxious people use social VR for social communication. There is a lack of in-depth research on how individuals with SAD use social VR for their social needs. In this paper, we report the result of a qualitative analysis of self-reported socially anxious individuals' experience in social VR, an emerging technology that provides new, embodiment, and immersive interactive virtual experiences, in Reddit (public online discussion) comments and posts.

3 METHOD

Data collection. We focus on first-person and narrative accounts of socially anxious people's experiences using various social VR applications to have social interactions. Many users highly engage in popular online forums to share their experiences of technologies

Table 1: Frequency of social VR platforms name in posts and comments

Platform's name	Frequency
VR Chat	181
Rec Room	22
AltspaceVR	6
BigScreen	4
VTime	1
No specific name	136

with others who have similar experiences. Therefore, we collected data from Reddit by using Reddit search API ⁶ between July and November 2022. First, we used related keywords in the literature (e.g., SAD, social anxiety, fear of public speaking, social phobia) in 12 VR, social VR, and social anxiety-related subreddits on Reddit that we have found through investigation (e.g., r/VRChat)⁷. Second, after the first iteration, we added more keywords based on users' comments and posts (e.g., socially awkward, anxiety attack, social disorder). Totally, We found 2415 posts or comments with the searched keywords. Then, filtering out irrelevant posts (e.g., social anxiety in the offline world and planning to start social VR but not about the usage of social VR in overcoming SAD), we collected 351 Reddit rich posts and comments for further analysis. In table 1, the frequency of using various social VR by socially anxious people is shown.

Data Analysis. We conducted an iterative data collection and analysis. We used in-depth qualitative analysis of the collected data [33] by adopting a Grounded Theory approach [7]. After each iteration, the first author closely read through the collected data line by line to have a whole picture of socially anxious people's experience in social VR and gather more rich data until data was saturated. Then, the first author identified a set of initial themes emerging in the data that can be used to answer our RQs through open coding [7]. Finally, both authors iteratively and collaboratively synthesized and revised these themes through axial coding and focused coding [7] to provide a detailed description of the role of social VR in mitigating SAD.

4 FINDINGS

This section describes how do socially anxious individuals use social VR for social communication (RQ1). Then, we identified strategies users use to mitigate social anxiety (RQ2). Finally, we highlight how practicing social interaction in social VR will affect users' offline world communication (RQ3).

4.1 Ways to leverage Social VR to combat Social Anxiety

Our findings indicate that social VR, beyond just virtual social interaction for ordinary people, can be leveraged to create opportunities

⁶<https://redditsearchtool.com>

⁷Subreddits used in this research include: r/AltspaceVR, r/VRchat, r/bigscreen, r/HighFidelity, r/OculusQuest, r/oculus, r/RecRoom, r/RecRoomVR, r/SocialVR, r/vTime, r/socialanxiety, and r/socialanxietyfriends

for individuals suffering from SAD through social VR's unique affordances and friendly environment.

4.1.1 Unique Social VR Affordances Help Combat Social Anxiety. In social VR, users experience immersive and realistic communication through embodied avatars with full/partial body tracking, voice, and the ability to customize their self-presentation. Many users highlighted that the ability to customize their self-presentation through an avatar reduced or eliminated the fear associated with their real-world appearance. For example, a user shared: "[...]I have severe anxiety and the moment someone talks to me directly, I freeze up [...] i don't have to use an ugly default skin and have already purchased a nice model i like (I am constantly being made fun of for my awkward banana outfit).", the user appreciated the ability to conceal their anxiety signs and the unwilling offline world appearance behind the unhumanized avatar. The fun outfit not only helped to hide the true face but also allowed them to find cues to start a conversation with others. A shared comment also admired anonymity under the shield of the avatar: "I tend to have social anxiety as well[and come off as quiet or shy around people who don't know me]. In RecRoom, it is anonymous enough that you don't have to worry about body language or appearance as much, but not anonymous enough for (most) people to become abusive. The fact that communication is voice based rather than typed messages [...]tends to keep people civil", in social VR boundary between the offline and online worlds are blurred. However, people are physically present in social VR through a virtual presentation, which allows them to hide their shyness behind the avatar and not worry about physical signs of being anxious while communicating verbally.

According to our data, several users shared that features such as joining diverse environments and having the ability to mute, block, or leave, gave them the power to manage uncomfortable situations. In the offline world, running away from an unpleasant situation may not be a feasible option for dealing with anxiety in offline world, but it is a viable approach in social VR. Several users on highly praised these technique as a means of managing their anxiety. For example, a user commented how social VR was safer for social communication compared to other online games, emphasized that undesirable behavior is manageable by muting the person or quickly removing themselves from the situation: "I just got into rec room this past week.[...] I'm generally anxious about social stuff. [...] There's so much to do and for the most part not a ton of the toxicity I've come to expect from online games. Occasionally I've come across people being jerks to kids [...]but generally if you call it out they stop and if not it's easy enough to mute and move on.". Another user shared the same story: "Coming from someone who has a bit of social anxiety, one big thing that helped me is just remember you can always leave, block, or mute people. If there's a situation you feel awkward in join another world and you will most likely never have to see those people again.", for this user, transferring their avatar to another location or muting to no longer hear the voice of the person who caused an inconvenient condition was helpful. Another social VR benefit that some users mentioned were the ability to choose a virtual space based on the level of the crowd, which eases starting social interactions: "I have massive social anxiety, [...]Thankfully, I've been able to join up with a real chill group that makes the game and experiences fun. Not so much when it's a room of like 30-40 people,

but the sub-15 rooms are the best.", and "[...]You can just hang out in a private room for awhile until you feel comfortable to jump into a public room.", the variety of virtual environments in social VR allowed these users to pick the place that felt more comfortable. From these comments, it can be inferred that for both users, the ability to select a setting with a specific number of people present was an opportunity to ease the interaction.

4.1.2 Engaging in A Safe and Friendly Space to communicate. It is noteworthy that, as Dane mentioned in his research [1], the diversity of users in social VR makes a supportive, safe, and comfortable place for LGBTQ to come out to make a social circle, also applied to individuals with SAD. People suffering from SAD also believed that social VR is a safe space with friendly people to start social interaction. A user described how other users reached out to the user and asked to join their groups or start a communication helped the user to be involved in an interaction with low social skills: "[...]It was really strange in the first couple weeks to go online and be invited into worlds to hang out with folks or to have a friend join on me and ask 'How you been? [...], and I'm very thankful for having met so many amazing people." supporting mute users and involving them in a conversation was appreciated by a user as it was a starting point for them: "[...]when I first got my headset, I just kinda dropped into public worlds until I found some interesting people and just listened. [...] Sometimes the people kinda pulled me into the conversation themselves, and it helped me open up a bit.", in this story, the users had a chance to be in a public place to see more people while being quiet was not awkward as in the offline world. Therefore, the user has time to explore until they encounter a friendly group of people who try to involve the user in a conversation. Another user shared a story regarding a chain of empathy and tolerance: "I've met some great people doing this that I otherwise would have never gotten the chance to talk to! I started out the same way, [...] but now I'm far more sociable[...]. One of the first things I do is find people who are by themselves and seem hesitant, I start talking to them and get em warmed up and then try and lead them to a group in an attempt to expose them to a few more people.", interestingly, this user, who suffered from SAD and achieved social skills through social VR, sought to create a chain of empathy and ease the process of initiating interactions for other individuals with SAD. Some users' reflections about their experience differed from others: "[...]I have pretty bad social anxiety myself, and Rec Room has definitely helped that so much. I've found that a lot of people who play Rec and stick around tend to have some sort of anxiety.", and "Huge percent of people have anxiety that plays vrchat. That's the beautiful part of an avatar and an alias.", from these users' overviews, many people in social VR had anxiety, which was the reason for empathy and tolerance for others, that led to a friendly environment.

4.2 Strategies users used to mitigate social anxiety

As our analysis of Reddit posts and comments revealed several strategies that users used to overcome their social anxiety through social VR.

4.2.1 Keep doing small steps continuously. In our data, users commented that they started with small steps and gradually built upon

them until they had the confidence to take on bigger challenges. This gradual approach of building upon small successes was seen as a key strategy for improving their ability to interact with others. One post described: *"Tell yourself you'll try out a room for 30 seconds, and if you don't like how it feels, you can take your headset off and just not try it out for another while."*, one of the small steps for this user was to experience entering a room for several seconds to verify how comfortable the room would be for seeking interaction and they stuck to this strategy until they found a room that made them less anxious. First move for some Reddit's users were to import techniques from the real world to manage anxiety in social VR: *"[...]Similar to IRL anxiety, I've found that going to the same places, over time I become familiar with that place and become comfortable being there."*, and *"[...]An extrovert friend once told me a technique which kind of changed my life and allowed me to get conversations going. I even met my current wife using it, so if that's not an endorsement, idk what is. [...] try to write down 10 to 15 questions you think people will like to talk about and memorize them. It's a great technique for IRL social situations too, and through practice you'll learn that it's really not hard to keep conversation going[...]"*, practicing this tiny step not only helped the user gradually achieve social skills but also assisted made an emotional connection to find the loved one. For another user, using imagination instead of reality was a start step: *"[...]what i did was i started to think that everything i saw in my headset was just a video game and the people in it isn't real but once you make good friends with someone, start to think that they are real people"*, this user considered that ignore the fact of interacting with real people and involved imagination with reality to make the situation controllable. As the user could not control the other users, they tried to make them under their control by separating them from reality.

4.2.2 Nonverbal communication to Break the Ice. Nonverbal communication in social VR is similar to face-to-face communication in terms of hand gestures and facial expressions, but it is not as awkward as it is in the offline world. This is because it is an accepted form of interaction for various reasons, such as privacy concerns [21] and LGBTQ individuals avoiding speaking [1]. A user commented nonverbal communication with embodied interaction through an avatar was a strategy for them to start interaction, as it gave the user time to prepare for communicating verbally: *"[I]did this. Found a cute avatar and just people watched using the emojis/reactions to respond to people. After a few hours I got enough courage to join a group and talk to them."*. Another user also shared: *"[...] Nonverbal communication is a very pure form of interaction, and is an interesting way of getting to know people in VRChat. Likewise, if you run into a group of friends that seem cool, you could hang out with them as a partial mute, and mainly communicate thru hand gestures and facial expressions. Most every group of friends on VRC has one or more mutes"*, this user not only has used mute interaction as a strategy but also believed it as a deep and meaningful form of communication. In the user's opinion, nonverbal communication was a potential situation to get familiar with people and the environment to increase confidence to start verbal communication. Another user reported a successful experience: *"[...] I've been staying mute in VRC for almost 2 years to my social anxiety and yes I have a lot of close friends in VRC but 3 months ago I decided to face my fear which*

I did it. Now I really do enjoy talking with them normally.", by using nonverbal communication as a technique for about two years, their confidence in facing their anxiety was boosted, and they gradually started to talk.

4.3 Influences of virtual social interaction on the offline social life

Our analysis of collected data has revealed that social VR not only presents an avenue for individuals with SAD to engage in virtual communication but also positively affects their offline social world. First, according to users' comments social VR can compensate for a lack of social interaction in the real world and can lead to feelings of contentment and well-being. For example, a user reported: *"my social anxiety is horrible and for years until i got on vrchat[...]. i got on vrchat and quickly made so many friends even though im a mute, [...]i feel really happy to be alive right now."*, in this story, the user mentioned there are opportunities to make social relationships even with low social interaction, nonverbal communication, which led to the user's well-being. Another user also shared that the usage of social VR led to their happiness: *"I'm an introvert [...]I log into VRChat every single day, even if only for an hour some days. My social skills have improved a lot, my anxiety has decreased a decent amount, my depression has been close to non-existent the past 5-6 months."*, as a result of socializing through social VR, the user's confidence was boosted, and their anxiety and depression were decreased. A user shared how they surprised their psychiatrist and therapist.

"During Lockdown, I told my Psychiatrist about how being immersed with online friends made me feel much less isolated. It also improved [...]to understand subtle verbal cues. [...]He [...]told me to log on every day, if I can. My Therapist has noticed things too. I have social flaws. [...] VR Chat allowed me to see and understand my social flaws, due to being able to safely experience them often. My therapist then used that understanding to get to the root causes of them, and now those flaws are mostly gone."

This user reported that social VR helped them better understand people's body language, expressions, and tone of voice, which prevented misunderstandings, and their improvement was evident for their psychiatrist and therapist. They found that exploring different VR environments with various people was more enjoyable than clinic therapy and helped accelerate their treatment by having the opportunity to experience different social interactions in a safe space repeatedly.

Second, the collected data revealed that the enhancement of social abilities was not restricted to social VR environments; it extended to users' social interactions in offline settings. For example some users shared: *"Even though irl I'm really socially awkward and barely talk to new people in the half year I've been playing vrchat I've been able to talk more and more to new people even irl and now I got to start college without knowing anyone pretty easily because I got used to talking to new people."*, and *"[...]As someone with severe social anxiety I was able to learn how to socialize because of this damn game. [...]Fast forward a year and half. Not only can I hold a proper conversation in VRchat, but that skill has transferred over to real life. [...] I'm still awkward as fuck but I've learned to live with it,"*

although both users still counted themselves as a socially awkward, but both admitted that practice of social skills within social VR facilitated the transfer of these skills to real-world. Another user reported the same story: *"I started VRChat with severe social anxiety back in December and I'm so much better now, even out of VRChat I've gained a lot more confidence and can talk to people."* In the above stories, the users admitted that communication in the offline world is still difficult for them; however, using social VR and practicing social communication gave them more acceptance of themselves as they learned how to deal with their anxiety in the offline world. Another user raised: *"I wasn't very social and had trouble making friends. VRChat taught me how easy it is to approach someone, i got to practice having conversations with strangers[...] This seriously helped me irl and now I'm less hesitant to meet new people."* According to this user's comment, social VR allowed users to practice social interaction in a different social environment which helped their social skills improve.

5 DISCUSSION

Similar to how individuals with SAD may engage in traditional forms of social networking, such as text- or video-based platforms, to compensate for the lack of face-to-face interactions, our findings also confirm that people with SAD use social VR for social interaction. However, social VR with immersive and embodied features, and high degree of resemblance of offline world activities can covers the barrier of face-to-face interaction in traditional social networks and online gaming. In addition, Our study highlights the potential of social VR to address the challenges faced by traditional exposure therapy and VRET because of the variety of prepared social situations with users' control over situations. Our research has highlighted how socially anxious individuals use social VR for their social needs (RQ1), the strategies that they use to practice social interaction in virtual environments (RQ2), and how practicing virtual social communication can influence their offline social lives (RQ3). The implications of this research are to better understand the impact of social VR communication on individuals with SAD, and to inform the design of more socially supportive and satisfying VR platforms in the future.

5.1 Novel Online Social Spaces as an opportunity For People with SAD

Prior literature has shown that traditional clinical treatments for SAD may have barriers to effectively recreating interpersonal situations, as well as limitations in personalization for each individual. Additionally, individuals with SAD may not desire to be identified as having the disorder or may not have access to treatment, making traditional exposure therapy difficult to access [25, 29]. While VRET addresses some of these barriers by mimicking social conditions in a virtual environment, it can be time-consuming and costly to personalize for each individual [8, 10]. Moreover, online communication platforms such as online gaming and social networks enable individuals with SAD to communicate through text with others, but excessive use of these platforms can also raise levels of anxiety due to the limitations of face-to-face communication [5, 6, 13, 17, 23, 28, 34]. Our findings highlighted three ways in which social VR can uniquely support individuals with SAD

in meeting their social needs and experiences. First, social VR in somehow addresses limitations of online platforms and traditional and contemporary exposure therapy. One unique aspect of social VR spaces is the sense of embodiment (i.e., experiencing a virtual body as a representation of our own body in a virtual environment [31]) through partial or full-body tracked avatars, and more customized avatar design. On the one hand, the fear of social situations due to scrutiny, negative evaluation by others, and physical symptoms such as shaking, blushing, and appearing anxious in social communication, can be concealed behind the avatar. While the experiencing social interaction is more realistic as the users are physically involved situation, and it is not just "looking at desktop" as they used to in traditional CMC. On the other hand, individuals can be involved in virtual social interaction while not being identifiable as having SAD, which puts less pressure on the user.

Second, social VR replicates mundane everyday activities in a highly realistic simulated 3D virtual environment in embodied and immersive ways, similar to face-to-face communication. The similarity of social VR to real-life interaction helps users experience social interaction in a more realistic way, which can solve the lack of face-to-face interaction in online communication platforms. In contrast to VRET and traditional exposure therapy, social VR does not need to recreate interpersonal situations to expose the user because there are pre-made various virtual social conditions that users would be exposed to. Therefore, there is no need to ask users to imagine the feared situation because those situations can naturally happen to users in the virtual society. In addition, users have control of undesired situations through social VR features (e.g., blocking, muting, leaving, and taking off the headsets), which gives them the confidence to experience social interaction in social VR.

Third, most of the people in social VR seek to socialize with other people all around the world. Based on users' reports, the nature of social VR, the diversity of people, accepting non-verbal communication as a natural way of interaction, and the presence of individuals with low social interaction skills in social VR led to the formation of a supportive community that involves even silent users in the interaction, which could serve as a starting point for social interaction.

A summary of our findings is provided in the following: (1) emphasize the unique advantages of leveraging social VR to combat social anxiety compared to traditional social networking sites, online games, clinical exposure therapy, and VRET; (2) highlight how the diversity of features and activities helped socially anxious users with different levels of social anxiety to be able to involve in social interaction through employing personalized strategies to leverage social skills; (3) shed light on a positive effect of social VR on people's offline world, their experience of well-being in daily life, and the possibility of transferring achieved virtual social skills to the real world.

5.2 Limitation

Our study mainly draws on Reddit posts, leading to a potential bias toward social VR users who are also active Reddit users. Since Reddit is an anonymous forum, verifying users' demographic information is challenging. Also, the data lacked users' anxiety levels, specified situations that make users anxious, and the duration of using social

VR for this purpose. We plan to conduct semi-structured in-depth interviews and a large-scale survey with a broader participant pool, diverse demographics, and various levels of social anxiety to investigate the difference in the experience level of anxiety in the offline world and social VR. We also plan to explore people's unique strategies, challenges, and expectations for using social VR to mitigate social anxiety.

6 CONCLUSIONS

How would an online virtual society resembling offline world action and activities benefit people suffering from social anxiety? This paper has explored characteristics of social VR, including embodiment interaction through an avatar, blocking, muting people with undesired behavior, and leaving a world. Those help people with social interaction disorder find these virtual social environments safe places to move out and practice social skills. Social VR affordances distinguish it from traditional (e.g., text- or video-based) social networking sites and online games in which users merely look at a computer screen for interaction. Practicing social skills with full-body or partial-body tracked avatars with embodied interaction similar to face-to-face communication and a wide range of activities boosts socially anxious people's confidence to transfer their social skills to the offline world. As the metaverse paradigm continues to expand and transform social interactions, we hope the knowledge gained from these insights will guide future research and design endeavors in developing social VR environments that are safer and more supportive for individuals experiencing social anxiety.

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